Foizi va qismiga ko'ra sonning o'zini topish

Chekda faqat chegirmali narx yozilgan. Asl narx qancha edi?

1-QISM

Mavzu

ESLAB OLAYLIK

- 1 butun 100% ga teng.
- Kasrni foizga aylantirish: $\frac{9}{20} = \frac{45}{100} = 45\%$.
- Sonning foizi: 640 ning 15% i — $640 : 100 \cdot 15 = 96$.

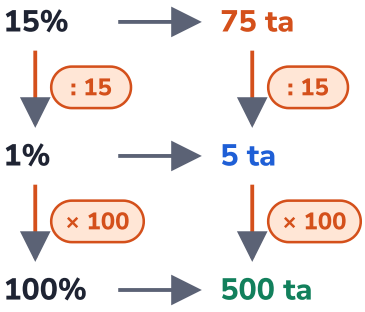
1-QOIDA · SONNING FOIZIGA KO'RA O'ZINI TOPISH

Sonning foizi ma'lum bo'lsa, avval sonning 1% ini, keyin 100% ini toping:

- Sonning 36% i $= 252$
- Sonning 1% i $= 252 : 36 = 7$
- Sonning 100% i $= 7 \cdot 100 = 700$

1-MISOL · AVTOTURARGOH

Avtoturargohdagi mashinalarning 15% i qizil, ular 75 ta. Avtoturargohda nechta mashina bor?



1. 1% — $75 : 15 = 5$ ta.
2. 100% — $5 \cdot 100 = 500$ ta.

2-MISOL · SON

Sonning 20% i 46 ga teng. Sonni toping.

1. 1% — $46 : 20 = 2{,}3$.
2. 100% — $2{,}3 \cdot 100 = 230$.
3. Tekshirish: 230 ning 20% i — $230 : 100 \cdot 20 = 46$.



Chekda faqat QQS yozilgan. Narxni shundan topsa bo'ladimi?

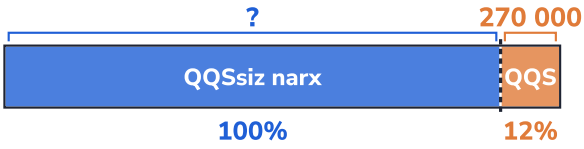
2-QOIDA · QQS

QQS (qo'shimcha qiymat solig'i) — narxga qo'shiladigan soliq.

QQS dastlabki narxning 12% iga teng.

1-MISOL · QQS

Planshet chekida «QQS: 270 000 so'm» deb yozilgan. Planshetning QQSsiz narxi qancha?



1. 12% — 270 000 so'm; 1% — $270\,000 : 12 = 22\,500$ so'm.
2. 100% — $22\,500 \cdot 100 = 2\,250\,000$ so'm.

2-MISOL · OMONAT FOIZI

Omonat — bankka qo'yilgan pul. Yillik foiz 14%. Bir yilda 980 000 so'm omonat foizi olindi. Bankka qancha pul qo'yilgan?

Bir yilda olingan omonat foizi — yillik foiz, qo'yilgan pul — 100%.

1. 14% — 980 000 so'm; 1% — $980\,000 : 14 = 70\,000$ so'm.
2. 100% — $70\,000 \cdot 100 = 7\,000\,000$ so'm.

25% chegirma bo'lsa, narxning qanchasini to'laymiz?



3-QOIDA · CHEGIRMADAN OLDINGI NARX

20% chegirmada: $100\% - 20\% = 80\%$. Chegirmadan keyingi narx dastlabki narxning 80% iga teng.

25% chegirmada esa $100\% - 25\% = 75\%$: chegirmali narx dastlabki narxning 75% i.

1-MISOL · DASTLABKI NARX

Ryukzak 25% chegirma bilan 156 000 so'mga sotildi. Uning chegirmasiz narxi qancha edi?



1. To'langan narx — $100\% - 25\% = 75\%$.
2. 1% — $156\,000 : 75 = 2080$ so'm.
3. 100% — $2080 \cdot 100 = 208\,000$ so'm.

2-MISOL · CHEGIRMA MIQDORI

Velosiped 35% chegirma bilan 715 000 so'mga sotildi. Chegirma necha so'm bo'lgan?

1. To'langan narx — $100\% - 35\% = 65\%$; 1% — $715\,000 : 65 = 11\,000$ so'm.
2. Chegirma — 35%, ya'ni $11\,000 \cdot 35 = 385\,000$ so'm.

KASR QISMI BERILSA

Kasr qismini foizga aylantiring, keyin qolgan qismning foizini toping.

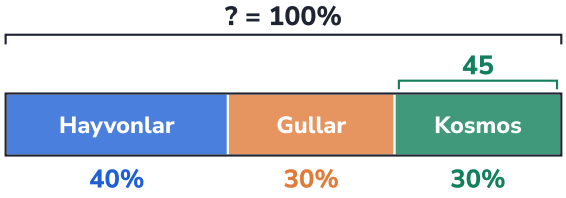
1-MISOL · SONNING KASR QISMI

Sonning $\frac{3}{4}$ qismi 150 ga teng. Sonni toping.

1. Foizga aylantirish: $\frac{3}{4} = \frac{75}{100} = 75\%$.
2. $1\% — 150 : 75 = 2$; $100\% — 2 \cdot 100 = 200$.
3. Bo'laklar bilan ham bo'ladi: son — 4 bo'lak, 3 bo'lagi 150 ga teng. 1 bo'lak — $150 : 3 = 50$; son — $50 \cdot 4 = 200$.

2-MISOL · KASR, FOIZ VA QOLGAN QISM

Kolleksiyadagi markalarning $\frac{2}{5}$ qismida hayvonlar, 30% ida gullar tasvirlangan, qolgan 45 tasi — kosmos haqida. Kolleksiyada nechta marka bor?



1. Hayvonlar: $\frac{2}{5} = \frac{40}{100} = 40\%$.
2. Kosmos: $100\% - 40\% - 30\% = 30\%$ — 45 ta.
3. $1\% — 45 : 30 = 1{,}5$; $100\% — 1{,}5 \cdot 100 = 150$ ta.

2-QISM

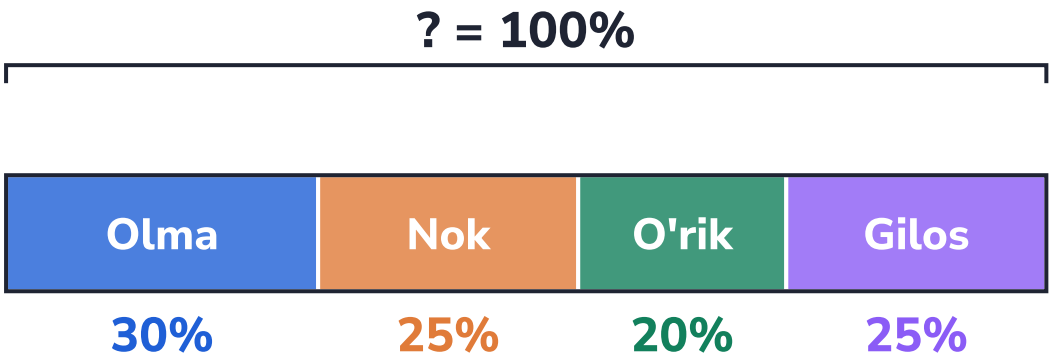
Misollar ishlash



Berilgan son butunning necha foizi? Masalani shu savoldan boshlang.

1-MISOL · IKKI BO'LAK BIRGA

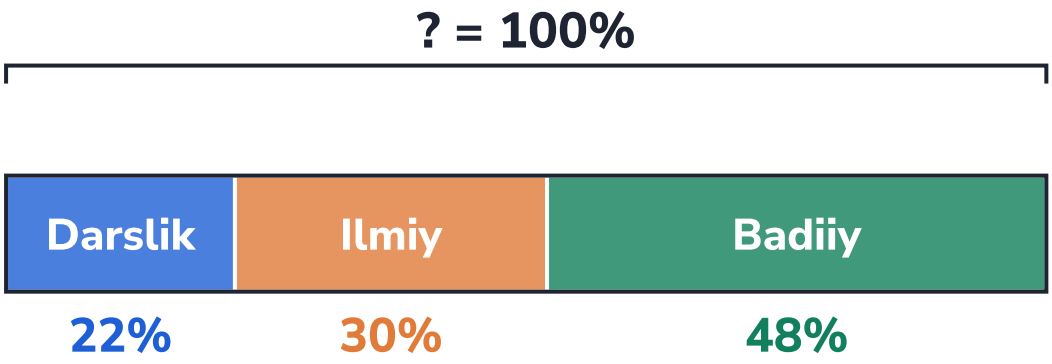
Tasmada bog'dagi mevali daraxtlar ko'rsatilgan. O'rik va gilos daraxtlari jami 270 tup. a) Bog'da necha tup daraxt bor? b) Olma va nok daraxtlari jami necha tup?



1. O'rik va gilos: $20\% + 25\% = 45\%$ — 270 tup.
2. 1% — $270 : 45 = 6$ tup.
3. a) 100% — $6 \cdot 100 = 600$ tup.
4. b) Olma va nok: $30\% + 25\% = 55\%$, ya'ni $6 \cdot 55 = 330$ tup.

2-MISOL · FARQ BERILGAN

Maktab kutubxonasiga yangi kitoblar keldi. Ularning 22% i darslik, 30% i ilmiy kitob, qolganlari badiiy. Badiiy kitoblar darsliklardan 52 ta ortiq. a) Kutubxonaga hammasi bo'lib nechta kitob keldi? b) Badiiy kitoblar nechta?

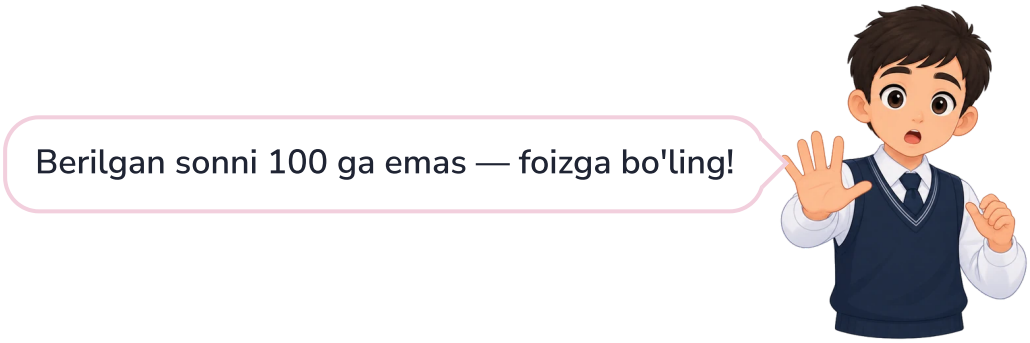


1. Badiiy kitoblar: $100\% - 22\% - 30\% = 48\%$.
2. Farq: $48\% - 22\% = 26\%$ — 52 ta.
3. 1% — $52 : 26 = 2$ ta; a) 100% — $2 \cdot 100 = 200$ ta.
4. b) Badiiy kitoblar 48%, ya'ni $2 \cdot 48 = 96$ ta.

3-MISOL · QOLGAN ISH

Brigada paxta terish rejasining 52% ini bajardi, yana 96 t paxta terishi kerak. a) Reja necha tonna? b) Necha tonna paxta terildi?

1. 96 t — qolgan ish: $100\% - 52\% = 48\%$. *(96 t terilgan emas, qolgan qism.)*
2. 1% — $96 : 48 = 2$ t.
3. a) 100% — $2 \cdot 100 = 200$ t.
4. b) Terildi: $200 - 96 = 104$ t.



DIQQAT!

1. Sonning 40% i 64 ga teng:
   ✗ son $= 64 : 100 \cdot 40 = 25{,}6$ → ✓ **1% — $64 : 40 = 1{,}6$; son — $1{,}6 \cdot 100 = 160$**
2. 1% kasr son chiqishi mumkin (1,6) — bu oraliq son, javob emas.
3. 15% chegirma, chegirmali narx 85 000 so'm:
   ✗ 85 000 — bu 15% → ✓ **85 000 — bu 85%; asl narx $85\,000 : 85 \cdot 100 = 100\,000$ so'm**
4. $\frac{1}{4}$ qismi va 30%:
   ✗ $\frac{1}{4} + 30\% = 31\%$ → ✓ $25\% + 30\% = 55\%$
5. QQS bilan birga narx 100% emas, 112%: QQSsiz narx — 100%, QQS — 12%. Dazmolning QQS bilan narxi 403 200 so'm bo'lsa: 1% — $403\,200 : 112 = 3600$ so'm, QQSsiz narx — $3600 \cdot 100 = 360\,000$ so'm.

3-QISM

Mustahkamlash

1. Sonni toping.

   a) 20% i — 34 **b)** 50% i — 47 **c)** 25% i — 19

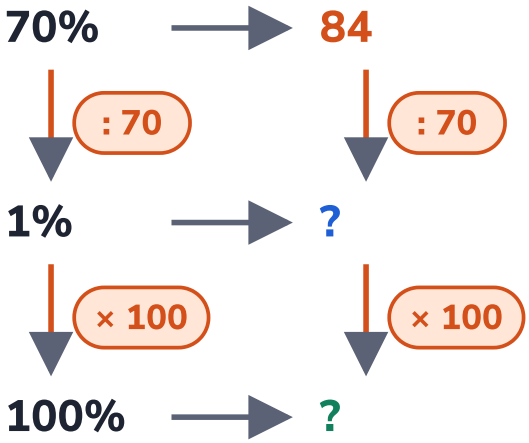
   d) 1% i — 8

2. Avval 1% ni, keyin sonni toping.

   a) 12% i — 54 **b)** 35% i — 91 **c)** 64% i — 112

   d) 8% i — 22

3. Foiz zinasidagi «?» o'rniga sonlarni yozing.



4. Partiyadagi lampochkalarning 2% i nosoz chiqdi — 7 ta. Partiyada nechta lampochka bor?

5. Stol lampasining chekida «QQS (12%): 21 600 so'm» deb yozilgan. Lampaning QQSsiz narxi va QQS bilan narxi qancha?

6. Yillik foiz 16%. Bir yilda 1 360 000 so'm omonat foizi olindi. Bankka qancha pul qo'yilgan?

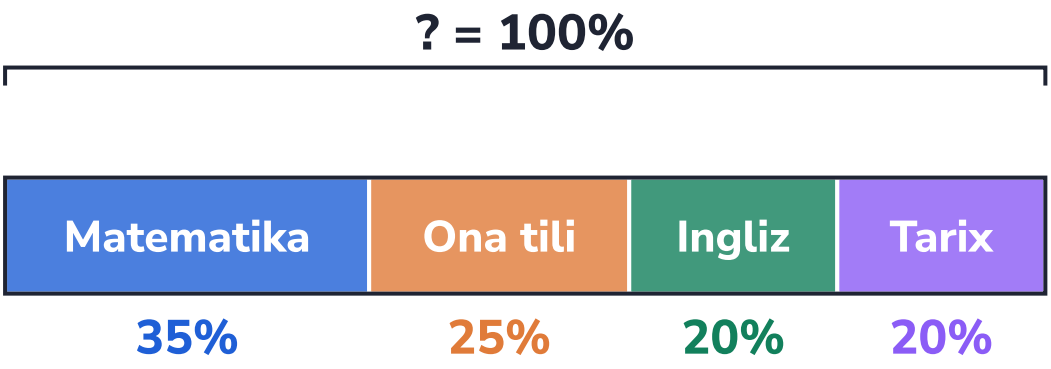
7. Do'kon hamma o'yinchoqlarga 15% chegirma e'lon qildi. Konstruktorning yangi narxi 161 500 so'm. Uning eski narxi qancha edi?

8. Sonning 45% i 81 ga teng. a) Sonni toping. b) Sonning 20% i nechaga teng?

9. Gulzordagi gullarning 44% i lola, 32% i atirgul, qolgan 108 tasi nargis. Gulzorda nechta gul bor?
10. Sayohatchilarning $\frac{1}{4}$ qismi avtobusda, 40% i poyezdda, qolgan 70 nafari samolyotda keldi. Jami nechta sayohatchi keldi?
11. Lagerdagi bolalarning $\frac{3}{20}$ qismi suzish, 40% i futbol, qolganlari shaxmat to'garagiga qatnaydi. Futbolga qatnaydiganlar suzishga qatnaydiganlardan 50 nafar ko'p. Lagerda nechta bola bor? Shaxmatga nechta bola qatnaydi?

Matnli masalalar

12. Telefon batareyasining 10% i 12 daqiqa ishlashga yetadi. To'la batareya necha daqiqaga yetadi?
13. Basketbolchi tashlagan to'plarining 75% ini savatga tushirdi — 36 ta. U nechta to'p tashlagan? Nechtasi tushmagan?
14. Bo'yoqchi devorning 25% ini bo'yadi, yana 48 m² bo'yashi kerak. Devorning yuzi necha m²?
15. Tasmada sinf o'quvchilarining sevimli fanlari ko'rsatilgan. Ona tili va ingliz tilini tanlaganlar jami 18 nafar. a) Sinfda nechta o'quvchi bor? b) Matematikani nechta o'quvchi tanlagan?



4-QISM

Mustaqil ish

1. Qaysi sonning 30% i 39 ga teng?
2. Termos chekida «QQS: 9 600 so'm» deb yozilgan. Termosning QQSsiz narxi qancha?

3. Yillik foiz 13%. Bank bir yilda 455 000 so'm omonat foizi berdi. Bankka qancha pul qo'yilgan?
4. Mebel do'konida divanga 40% chegirma qilindi va xaridor 690 000 so'm to'ladi. Xaridor chegirma tufayli necha so'm tejadi?
5. **Xatoni toping.** Do'konda 10% chegirma bor. Chegirmali narx 90 000 so'm. Nodir shunday yozdi: «Asl narx $= 90\,000 + 9\,000 = 99\,000$ so'm». Nodirning xatosi nimada? To'g'ri javobni toping.
6. **Qiyinroq.** Javondagi kitoblarning 40% i darsliklar, qolganlari ertak va she'r kitoblari. Ertak kitoblari she'r kitoblaridan 3 marta ko'p va 24 taga ko'p. Javonda nechta kitob bor? *(She'r kitoblarini 1 bo'lak, ertak kitoblarini 3 bo'lak deb oling.)*

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $34 : 20 \cdot 100 = 170$; b) $47 : 50 \cdot 100 = 94$; c) $19 : 25 \cdot 100 = 76$; d) $8 \cdot 100 = 800$.

2-masala

a) 1% — $54 : 12 = 4{,}5$; son 450. b) 1% — $91 : 35 = 2{,}6$; son 260. c) 1% — $112 : 64 = 1{,}75$; son 175. d) 1% — $22 : 8 = 2{,}75$; son 275.

3-masala

1% — $84 : 70 = 1{,}2$; 100% — $1{,}2 \cdot 100 = 120$.

4-masala

1% — $7 : 2 = 3{,}5$; 100% — $3{,}5 \cdot 100 = 350$ ta.

5-masala

1% — $21\,600 : 12 = 1800$ so'm; QQSsiz narx $1800 \cdot 100 = 180\,000$ so'm; QQS bilan $180\,000 + 21\,600 = 201\,600$ so'm.

6-masala

16% — 1 360 000 so'm; 1% — $1\,360\,000 : 16 = 85\,000$ so'm; qo'yilgan pul $85\,000 \cdot 100 = 8\,500\,000$ so'm.

7-masala

Yangi narx — $100\% - 15\% = 85\%$; 1% — $161\,500 : 85 = 1900$ so'm; eski narx $1900 \cdot 100 = 190\,000$ so'm.

8-masala

1% — $81 : 45 = 1{,}8$. a) $1{,}8 \cdot 100 = 180$. b) $1{,}8 \cdot 20 = 36$.

9-masala

Nargis: $100\% - 44\% - 32\% = 24\%$ — 108 ta; 1% — $108 : 24 = 4{,}5$; jami $4{,}5 \cdot 100 = 450$ ta.

10-masala

$\frac{1}{4} = 25\%$; samolyot: $100\% - 25\% - 40\% = 35\%$ — 70 nafar; 1% — $70 : 35 = 2$; jami 200 nafar.

11-masala

$\frac{3}{20} = \frac{15}{100} = 15\%$. Farq: $40\% - 15\% = 25\%$ — 50 nafar; 1% — $50 : 25 = 2$; jami 200 nafar. Shaxmat: $100\% - 15\% - 40\% = 45\%$, ya'ni $2 \cdot 45 = 90$ nafar.

12-masala

1% — $12 : 10 = 1{,}2$ daqiqa; 100% — $1{,}2 \cdot 100 = 120$ daqiqa (2 soat).

13-masala

1% — $36 : 75 = 0{,}48$; tashlagan $0{,}48 \cdot 100 = 48$ ta; tushmagan $48 - 36 = 12$ ta.

14-masala

48 m² — qolgan qism: $100\% - 25\% = 75\%$; 1% — $48 : 75 = 0{,}64$ m²; devor $0{,}64 \cdot 100 = 64$ m².

15-masala

Ona tili va ingliz tili: $25\% + 20\% = 45\%$ — 18 nafar; 1% — $18 : 45 = 0{,}4$. a) $0{,}4 \cdot 100 = 40$ nafar. b) $0{,}4 \cdot 35 = 14$ nafar.

Mustaqil ish

1-masala

1% — $39 : 30 = 1{,}3$; son $1{,}3 \cdot 100 = 130$.

2-masala

QQS — 12%; 1% — $9600 : 12 = 800$ so'm; QQSsiz narx $800 \cdot 100 = 80\,000$ so'm.

3-masala

13% — 455 000 so'm; 1% — $455\,000 : 13 = 35\,000$ so'm; qo'yilgan pul $35\,000 \cdot 100 = 3\,500\,000$ so'm.

4-masala

To'langan narx — $100\% - 40\% = 60\%$; 1% — $690\,000 : 60 = 11\,500$ so'm; chegirma — 40%, ya'ni $11\,500 \cdot 40 = 460\,000$ so'm.

5-masala

Chegirma asl narxdan olinadi, chegirmali narxdan emas. 90 000 so'm — asl narxning $100\% - 10\% = 90\%$ i; 1% — $90\,000 : 90 = 1000$ so'm; asl narx $1000 \cdot 100 = 100\,000$ so'm.

6-masala

She'r — 1 bo'lak, ertak — 3 bo'lak; farq 2 bo'lak — 24 ta, 1 bo'lak — 12 ta. Ertak 36 ta, she'r 12 ta, birga 48 ta — bu $100\% - 40\% = 60\%$. 1% — $48 : 60 = 0{,}8$; javonda $0{,}8 \cdot 100 = 80$ ta kitob.